An experiment investigating the effects of fat content on the amount of microbial growth in milk at varying temperatures.

IB Biology HL Internal Assessment

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1: Introduction

Bacterial spoilage within food is a widespread issue that affects the quality and safety of a variety of perishable food types. Dairy products, such as milk, are particularly susceptible to microorganism contamination due to a multitude of different factors including, but not limited to, inadequate milking, processing, storage, and transport conditions of raw milk (Calahorrano-Moreno et al., 2022). In an attempt to mitigate the adverse effects that microorganism contamination has on dairy products, I believe that it is important for the microbiological conditions that may or may not hinder bacterial growth to be researched.

The central focus of my Biology IA is to address the research question: "What effect does the amount of fat content have on the amount of microbial growth present in milk at multiple temperatures?" This question stems from the recognition that the nutritive components of milk, specifically fat, may impact the ability for microorganisms to grow and survive within the nutritionally dense beverage.

I am choosing to write my Biology IA on this topic because bacteria can react to the environment in a multitude of different ways, such as being able to quickly evolve drug resistance (Moreno-Gamez et al., 2015). However, from my own research, while I did find studies done on this specific topic/topics related to this, I did not find very many, making it an important area of study to explore further. It can be hypothesized that the fat content will help bacteria grow within milk since fat is made up of lipids which is a long-term energy source.

My IA seeks to contribute to the existing biological literature on food preservation by elucidating the dynamics of bacterial spoilage in milk at different fat levels. Understanding how fat content can affect the spread of bacteria within a food item holds practical implications for future food storage practices, such as through the optimization of dairy product storage by perhaps storing milk with different fat levels at different temperatures. Additionally, a deeper understanding of the relationship between fat content and bacterial spoilage can subsequently be used to improve food preservation practices within other food groups, such as within fruits, vegetables, and protein dishes.

2: Investigation

2.1: Hypotheses

H_1: Milk samples with higher fat content stored at both room and refrigerated temperature conditions will exhibit a significantly greater amount ($p < 0.05$) of microbial growth than those with lower fat content.

H_0: There will be no significant difference ($p > 0.05$) in the amount of microbial growth present at each fat level.

2.2: Background Information

Personal Engagement

Growing up, I have never been a big fan of milk. I, at least as of this point in my life, am not lactose-intolerant so I have no particular digestive issues when it comes to consuming the drink, but I have never liked the taste of it, albeit I have been a regular fan of other dairy products such as Greek Vanilla and Greek Mixed-Berry Yogurt created by Chobani. However, as an athlete, I am regularly exposed to milk in the fitness industry as it is a nutritious medium and contains plenty of calories. It was, for example, a staple in the diet of world-renowned bodybuilder Arnold Schwarzenegger, who thus caused it to become a popular staple among the community during and after his career. He was also an influence of mine to begin trying to drink milk more often.

Bacteria

Bacteria are a large group of single-celled microorganisms related phylogenetically that are often confused with but distinctly different from Archaea, a different group of prokaryotes. They are found in a wide range of shapes and sizes, such as rods and spirals. Bacteria are present everywhere: in the soil, water, extreme environments (Vreeland et al., 2000; Wanger et al., 2008), and even all over our own bodies; in the gastrointestinal tract alone, there are at least ten trillion microbes, composed of more than 1,000 species (Drexler, 2010).

Most of the bacteria in our bodies play beneficial roles, such as preventing the gut from being colonized by organisms that can cause disease, synthesizing vitamins, breaking down food into absorbable nutrients, and stimulating our immune systems (Drexler, 2010). However, this is not the case for every species of bacteria, as a small percentage of the world's bacteria cause infection and disease. Bacterial infections can be spread in many ways: through direct contact, the air, water, droplets, vectors (arthropods transporting pathogens from an infected host to an uninfected host by means of a bite), or a vehicular medium such as dairy products (Doron & Gorbach, 2018). Examples of bacterial infections transported by these means include Salmonella, Tuberculosis, and Lyme disease (Doron & Gorbach, 2018).

Microbial growth is heavily facilitated by environmental conditions, such as by the presence of nutrients and energy, as microbes need fats, carbohydrates, proteins, and others to survive, just like humans and animals do (Sargen, 2020). Other factors that can impact microbial growth are temperature, pH, amount of light, and moisture, all of which most types of bacteria have an optimal range to grow in.

Microbial growth in milk can be measured through a variety of methods, but the primary method involves a microscopic cell count. A microscopic cell count can be done after microbial staining techniques, which are commonly used in laboratories for classifying and identifying microorganisms. There are a number of different automated staining methods, such as the Grain stain, the Capsule stain, the Acid-fast stain, and the Endospore stain, but manual staining, such as with a methylene blue solution, remains the gold standard in this field (Siguenza et al., 2019). Manual staining is done by attaching a sample to a microscope glass, such as a hemocytometer (a modified slide used for cell counting), followed by staining the slide with a small amount of

dye to make it visible under a microscope (Siguenza et al., 2019), and the cells can then be counted individually. The number of microbes can be measured in CFU/mL, or colony forming units per millimeter of liquid, or cells/mL, depending on the procedure.

If a microscopic cell count of bacteria is done using a hemocytometer, it is important to note that there are two main types of cells: viable and nonviable cells. Viable cells are cells that are still metabolically or physiologically active, or in a word, alive (Fakruddin et al., 2013), whereas nonviable cells are not and are therefore themselves harmless, though their presence may still indicate potential microbial contamination by viable cells of the environment they were found in. During a bacteria cell count using a hemocytometer and methylene or trypan blue dye, nonviable cells will appear blue under light microscopy analysis because they are dead cells and the dye can therefore pass through the cell membrane, whereas it cannot in viable cells, making the two distinguishable by color (Fang & Trewyn, 2012). Viable cells are the important cells to pay attention to as they are alive and can pose real harm to consumers, however, in theory, an increase in the number of nonviable cells can indicate an increase in the death of viable cells within a milk sample, meaning that the number of nonviable cells can be used as an indirect metric to assess milk spoilage.

Milk

Milk is a white liquid produced by the glands found in mammals, and is the primary source of nutrition for young mammals, including humans through the process of breastfeeding. It remains an important component of a balanced diet as it contains many valuable constituents, primarily proteins and calcium, beneficial not only for their nutritive value but also for their biological properties, such as "...anticarcinogenic activities,", a carcinogen being a substance or organism capable of causing cancer, "...antihypertensive properties,", meaning it can help fight high blood pressure, "...immune system modulation, and other metabolic features," (FoodData Central, n.d.; Davoodi et al., 2016; Hutter, 2023).

The most common type of milk for humans to consume after infancy is cow's milk. The milk produced immediately after milking a cow is referred to as 'raw milk', which is milk that has not yet been purified to kill harmful bacteria, whereas pasteurized milk is raw milk that has been purified (CDC, 2017). The pasteurization of any food is the process of heating it to temperatures just below 100 degrees celsius, which extends its shell-life through the destruction of microorganisms and inactivation of enzymes (Fellows, 2009, p. 381-395). Common types of pasteurized milk include whole (full-fat) milk, 2% (reduced fat) milk, and skim (fat-free) milk; the primary difference between each being the amount of fat retained from the raw state. Microbial spoilage in milk can be easily identified by changes in color and texture, such as by the medium turning yellow and thick.

To prevent the microbial contamination of the medium, a variety of strategies are employed. Refrigeration, after the aforementioned method of pasteurization, stands as a primary practiced method, inhibiting microbial growth by maintaining low enough temperatures that slow down metabolic and reproductive processes within bacteria. Another method used is Ultra-High Temperature (UHT) processing, which is similar to pasteurization.

However, through UHT, milk is subjected to temperatures above 135 degrees celsius, further extending its shell-life by several months (Karlsson, et al., 2019). Additionally, food preservation can be done through the use of preservatives, which are substances added to foods to hinder microorganism contamination (B.L. Wedzicha et al., 1991). These substances can either be antimicrobial or antioxidants, inhibiting microbial growth or the oxidation of food parts (Sambu et al., 2022).

The nutrients found in each variant of pasteurized milk are as follows:

Hood **Whole Milk**	Hood **Reduced Fat Milk**	Hood **Fat-Free Milk**
		

Images of nutrition labels were taken by a Samsung Galaxy smartphone.

As shown above in each of the variants of pasteurized milk, there are plenty of nutrients that can be found within the liquid. Rich in multiple vitamins (notably B_{12} and D), minerals (notably calcium), fat (with the exception of the fat-free variant), and protein, milk provides an abundant source of nutrients that may support microorganism contamination through the promotion of metabolic processes.

Previous Studies

Zimmerman et al. (2001) aimed "...to evaluate the number of microbes and component degradation in raw milk,". The researchers used raw milk samples that were obtained from two different groups of cows: milk from cows that produced "high-fat" milk, and milk from cows that produced a "normal" amount of fat. Immediately after milking the cows, they transferred the milk samples to a dairy processing facility, filtered them, sampled them, and put them into 'Whirl-Pak' bags, storing them at either 39 or 45 degrees fahrenheit for up to eight days. However, contradictory to what I believe will occur, this study found that milk fat content did not affect bacterial counts, unlike other factors such as temperature and duration of storage; as both temperatures and duration of storage increased, greater bacterial growth was observed.

Yang et al. (2023) aimed, "...to determine how long [post-]expiration-pasteurized milk may still be safe to consume and what is the relationship between bacterial growth from the milk and time after expiration,". The researchers carried this experiment out by including milk with varying levels of fat content (3.25%, 2%, and 0%) and by incubating the samples with agarose gel, and then waiting to see how many days it took before each of the milk samples began to spoil. They found that there was no statistically significant difference between the amount of bacteria present on the first day after expiration and on the days following while each of the samples were refrigerated and unopened. However, within 24 hours of opening the milk cartons, they found that whole milk and skim milk samples which were six to eight or more days past their expiration date respectively began to show signs of bacterial spoilage (ex: sour taste, pungent smell), while the fat-free samples in the same conditions did not change within 10 days post-expiration (the maximum amount of days for this experiment), suggesting that the levels of bacterial growth increases when the fat content of the milk increases.

Since the findings for these studies were not very consistent with one another on the basis of differing fat affecting microbial growth in milk, I can expect to find either that they do play a role or that they do not play a role, and that other factors such as temperature and duration of storage may have a larger impact, though I will not be testing for the latter directly.

2.3: Variables

Independent variables: Fat Content (%), Temperature (°C)

Dependent variable: Amount of Microbes Present in Milk Samples (cells/mL)

Controlled variables		
Variable:	**Significance:**	**How will it be kept constant:**
Temperature	- This experiment will focus on bacterial spoilage under refrigerated temperatures, but an extra trial will be conducted at room temperatures to determine if there is a spoilage pattern. - Fluctuations in temperature can affect the rate at which microbial growth can occur, meaning both temperature conditions need to remain consistent throughout for accuracy.	In the refrigerated condition, the refrigerator will be set as close to 4 to 5 degrees celsius (39.2 to 41 degrees fahrenheit) as possible, which is about the most optimal temperature for most perishable food items. The temperature in both the refrigerator and room will be measured at the beginning, middle, and end of the experiment to ensure there were no major fluctuations, and an average temperature will be noted in the 'Control Variable Measurements' table in section 4.1 below.
Amount of Light	- The presence of light, whether artificial or sunlight, can either increase or decrease microbial growth (Lindell et al., 1995; Ward, 1984), so keeping it as consistent as possible will be necessary.	In both the refrigerated and room temperature conditions, there will be a constant light source for the duration of the experiment.
Milk Brand	- Different milk brands have slightly different nutritional constituents, which may or may not affect microbial growth. - Different milk brands also may have varying amounts of initial bacterial composition which could hinder the accuracy of the results.	All milk samples will come from the brand 'Hood'. This brand has all variations of milk that will be used in the experiment.
Initial Time of Milk Storage During Experiment → Consistent Time of Milk Purchase	- Storing the milk samples at different times may affect the total composition of bacteria within the samples, so a consistent initial storage time of each milk sample is needed.	At the time of storage, the time will be measured in 'hh:mm AM/PM' format and will be noted in the 'Control Variable Measurements' table in section 4.1 below.
Duration of Storage/Overall Use During Experiment	- For a similar reason of the initial time of milk storage, the duration of storage may also impact microbial growth.	The duration since purchase of each milk sample will be noted in the 'Control Variable Measurements' table in section 4.1 below.
Milk/Methylene	- Differences in milk volume, methylene	The amount of milk, methylene blue

Blue Dye/Water Concentration Per Solution	blue dye concentration, and water concentration may impact bacterial cell counts. Higher milk volume or methylene blue could increase bacteria counts, while water concentration variations could affect solute concentration and thus cell counts.	dye, and water will all accurately be measured and distributed into each solution using a micropipette.
Handling and Storage Procedures During Experiment	- Since bacteria is present in large quantities everywhere, it will be necessary to handle and store each of these milk samples with as little contact to the external environment as possible to minimize contact with external bacteria.	Through the use of certain Personal Protective Equipment (PPE), such as gloves and glasses, I will attempt to handle each sample with as much care as possible and expose them to as few bacteria-infected surfaces as possible.

Confounding Variables	
Variable:	**Possible effect:**
Milk Source Variability	Milk from different sources, whether or not from the same company, may contain inherent differences in the amount of microbes present, which may affect bacterial cell counts during the experiment.
Handling and Storage Prior to Purchase	During the handling and storage of each milk sample prior to purchase, exposure to contaminated surfaces, manufacturer and retailer handling, and storage temperatures all may play a role in subsequent bacterial growth during the experiment.
Initial Microbial Load Due to Transportation and Packaging Conditions	The possible presence of bacteria in the milk samples before purchase may make it more likely for further bacteria to grow in each sample; additionally, each sample may contain differing amounts of bacterial load, which may impact conclusions about fat level's effect on milk spoilage.
Time Since Milking	Milk samples obtained shortly after milking may contain fewer amounts of bacteria compared to samples that have been stored for longer periods, potentially influencing subsequent amounts of bacterial growth during the experiment.

3: Procedure

3.1: Materials/Apparatus

- **Milk Samples (2 per fat level; 2 per temperature condition) at ≈240mL:**
    - 4 Hood Whole Milk Samples.

    - 4 Hood Reduced Fat Milk Samples.
    - 4 Hood Fat-Free Milk Samples.
- **Sharpie and Tape:**
    - For milk sample and solution labeling.
- **12 ≈240mL (1 Cup; 8oz) Plastic Jars with Lids:**
    - For the storage of milk samples.
- **Refrigerator:**
    - For the storage of milk at refrigerated temperatures.
    - Temperature of Refrigerator: 4 to 5°C
- **Methylene Blue Dye:**
    - For cell staining.
    - Amount needed: 2µL.
- **Thermometer:**
    - For the monitoring and maintenance of temperatures during the experiment.
- **Microscope and Lens:**
    - For the microscopic cell counts.
    - Lens: 10x magnification with a 0.25 aperture.
- **Hemocytometer Pack:**
    - For more accurate manual cell counting than traditional microscopic slides.
    - Pack contents:
        - Hemocytometer (total volume of 10,000mL)
        - 2 Cover Slips
        - 4 Micropipettes
        - A 100-Page Lens Paper Book
        - Instruction Booklet

    - 0.2mL of Methylene Blue and Trypan Blue Dye

- **24 0.2mL (200µL) PCR Tubes:**
    - For diluted solutions.
    - 20 to 200µL Micropipette with Multiple Tips
    - For creating diluted solutions.
- **Personal Protective Equipment (PPE):**
    - For minimizing the amount of bacteria that may be transferred from the external environment, and for safety from microbial exposure.
    - Includes gloves for hand protection, glasses for eye protection, and antimicrobial wipes for cleaning certain equipment after use (ex: hemocytometer, cover slips).

3.2: Methodology

1. **Preparation of Milk Samples:**
    a. Record time of purchase (hh:mm AM/PM format).
    b. Label each milk sample based on its fat content (3.25%, 2%, 0%; 2 samples per fat level in 2 different temperature conditions).
    c. Pour ≈240mL of milk into each of the 12 jars and seal the lids.
    d. Store all samples at refrigerated temperatures for a couple hours before separating the samples into their respective conditions.
    e. After a couple of hours, store 6 samples at room temperature and the other 6 samples at refrigerated temperatures, and record the time they are stored (hh:mm AM/PM format).
    f. Leave the samples stored for 12 days.
2. **Microscopic Cell Count:**
    a. Use a micropipette with multiple pipette tips to distribute 8µL of milk from each sample, 2µL of methylene blue dye, and 90µL of distilled water into 12 different PCR tubes to create a 1:10 dilution, and shake thoroughly.

i. Take 10µL of this solution and add it to another 90µL of distilled water to create a 1:100 dilution for easier cell identification.

b. Prepare hemocytometer slide for microscopic cell count using stained samples of milk.

i. Use a micropipette to move a small amount of the solution to the hemocytometer, which can be applied through two different methods:

1. Capillary Action: Putting the micropipette up to the edge of the cover slip and pipetting a small amount of the solution to allow capillary action to suck the solution underneath the coverslip.

2. Manual Droplet: Pipette no larger than a pea-sized amount of the solution onto the upper/lower chamber of the hemocytometer and carefully place the cover slip on top for even spreading.

ii. Carefully place the hemocytometer onto the mechanical stage of the microscope, locking it in using the slide holder, and adjust the lens, light and contrast levels, and coarse and fine focus levels as needed for a clear view of the grid.

iii. Count microbial cells under the microscope with the hemocytometer slides and record data.

1. Non-viable cells will appear blue (see '2.2: Background Information' for explanation).

iv. Repeat steps 2a) → 2biii) for all solutions.

Image of microscope setup can be found in Appendix F.

3. Math Calculations:

a. Use data gathered from the bacterial cell counts to calculate the percentage of viable cells, the average number of nonviable cells, and the estimated concentration of the cells (measured in viable cells/mL).

4. Analysis:

a. Compare the cell counts in each milk sample with different fat content at both refrigerated and room temperatures to determine if there is a pattern between fat content and microbial growth.
    i. Use statistical analysis to assess the significance of the differences between each fat level ($p < 0.05$) for the cell counts in both the refrigerated and non-refrigerated samples.

5. **Conclusion:**
    a. Draw conclusions based on the data collected and data analysis.
    b. Relate the findings to the hypotheses and determine whether the original hypothesis can be sustained.
    c. Discuss the implication of fat content on microbial growth in milk and its relevance to future food preservation methods.

3.3: Risk Assessment

Safety Issues	**Microbial and Chemical Exposure** ● **Risk:** Potential exposure to bacteria while handling milk samples and methylene blue solution. ● **Mitigation:** Use of PPE (gloves and glasses).
Ethical Issues	There are no ethical considerations to be taken into account. External bacterial cultures were not used.
Environmental Issues	**Milk Sample Disposal** ● **Risk:** Improper disposal of milk samples can lead to potential environmental hazards. ● **Mitigation:** Diluting milk samples before disposal and cleaning storage containers thoroughly to prevent further contamination. **Chemical Disposal** ● **Risk:** Improper disposal of chemicals (methylene blue) can lead to potential environmental hazards. ● **Mitigation:** Further diluting the solutions with excess tap water and rinsing this down the drain (extra steps are needed if working with a more concentrated solution/solid form of the methylene blue dye).

4: Raw Data

An example for calculating each metric can be found in Appendix A.

Raw Data Table 1: Bacterial Growth Measured in Whole Milk (3.25%) Samples in Refrigerated and Room Temperature Conditions

Sample	*Percentage of Viable Cells	# of Nonviable Cells	Estimated Cell Concentration
4.5°C Sample 1	90.5%	9	21,500,000 viable cells/mL
4.5°C Sample 2	92.3%	1	3,000,000 viable cells/mL
22°C Sample 1	100%	0	5,000,000 viable cells/mL
22°C Sample 2	100%	0	1,500,000 viable cells/mL

**Rounded to 3.s.f.*
Example hemocytometer image of one grid of 4.5°C Sample 1 can be found in Appendix D.

Raw Data Table 2: Bacterial Growth Measured in Reduced Fat Milk (2%) Samples in Refrigerated and Room Temperature Conditions

Sample	Percentage of Viable Cells	# of Nonviable Cells	Estimated Cell Concentration
4.5°C Sample 1	91.7%	1	2,750,000 viable cells/mL
4.5°C Sample 2	88.9%	1	2,000,000 viable cells/mL
22°C Sample 1	100%	0	2,000,000 viable cells/mL
22°C Sample 2	100%	0	750,000 viable cells/mL

Raw Data Table 3: Bacterial Growth Measured in Fat-Free (0%) Milk Samples in Refrigerated and Room Temperature Conditions

Sample	Percentage of Viable Cells	# of Nonviable Cells	Estimated Cell Concentration
4.5°C Sample 1	85.7%	2	6,000,000 viable cells/mL
4.5°C Sample 2	85.7%	1	1,500,000 viable cells/mL
22°C Sample 1	50%	3	750,000 viable cells/mL
22°C Sample 2	N/A	0	0 viable cells/mL

4.1: Control Variable Measurements

Control Variable	Measurement
Average Temperature	Refrigerated Condition: 4.50°C Room Temperature Condition: 22°C
Milk Concentration	8μL per solution
Methylene Blue Dye Concentration	2μL per solution
Water Concentration	90μL per solution
Time of Milk Purchase	6:00 PM on March 6th, 2024
Total Time With Milk After Purchase + Total Duration of Experiment	Time after purchase until the end of exp. - 13 days (310h56m) Duration of experiment - 12 days (297h28m)

4.2: Qualitative Observations

- Samples left at room temperature had very different physical appearances depending on the amount of fat content present. For example, the 0% fat milk samples' white part, which is primarily made up of fat globules and proteins (Kitaguchi et al., 2012), had separated from the rest of the liquid, which did not happen in either the reduced fat or whole milk samples, albeit there were structural breakdowns in those themselves (see Appendix B).
- When creating the initial 1:10 diluted solutions, solution 1 from sample 1 of the 2% milk in the refrigerated condition appeared much lighter in color than the rest of the solutions, which were dark blue

(see Appendix C). Due to possible incorrect staining/varying amounts of liquids, a new solution from this specific sample was made and the solution that appeared lighter in color was not used for the cell count.

- While attempting to measure the solutions' bacteria cell counts from the initial 1:10 dilution, there appeared to be many bacteria cells present, however, the grid lines on the hemocytometer were too hard to make out, leading to a change from a 1:10 to a 1:100 dilution for easier grid line identification and thus cell counts (see Appendix E).
- Overall, the colors of the diluted solutions created from the room temperature milk samples were of a darker blue color than those of the refrigerated conditions, potentially indicating a greater amount of bacteria cells stained (see Appendix C), though this potential indication was not consistent with the data.

5: Processed Data

Processed Data Table 1: Percentage of Viable Bacteria Cells Across All Milk Samples (both Refrigerated and Room Temperature Conditions)

% of Viable Cells	Refrigerated Temperature	Room Temperature
Whole Milk	90.7%	100%
Reduced Fat Milk	90.5%	100%
Fat-Free Milk	85.7%	50%

Formula: $\frac{viable\ cells\ in\ RE\ sample\ 1 + viable\ cells\ in\ RE\ sample\ 2}{total\ number\ of\ cells\ across\ samples\ 1\ and\ 2}$

Example: Average percentage of viable cells in whole milk samples 1 and 2 = $\frac{86+12}{95+13}$ = *0.907 = 90.7%

**Rounded to 3.s.f.*

Processed Data Table 2: Average Number of Nonviable Bacteria Cells Across All Milk Samples (both Refrigerated and Room Temperature Conditions)

Avg. # of Nonviable Cells	**Refrigerated Temperature**	**Room Temperature**
Whole Milk	5	0
Reduced Fat Milk	1	0
Fat-Free Milk	1.5	1.5

Formula: $\frac{\#\ of\ nonviable\ cells\ in\ RE\ sample\ 1 + \#\ of\ nonviable\ cells\ in\ RE\ sample\ 2}{2}$

Example: Average number of nonviable bacteria cells in whole milk samples 1 and 2 = $\frac{9+1}{2} = 5$

Processed Data Table 3: Estimated Cell Concentration Across All Milk Samples (both Refrigerated and Room Temperature Conditions)

Avg. # of Nonviable Cells	**Refrigerated Temperature**	**Room Temperature**
Whole Milk	12,250,000	3,250,000
Reduced Fat Milk	2,375,000	1,375,000
Fat-Free Milk	3,750,000	750,000

Formula: $\frac{estimated\ cell\ concentration\ in\ RE\ sample\ 1 + estimated\ cell\ concentration\ in\ RE\ sample\ 2}{2}$

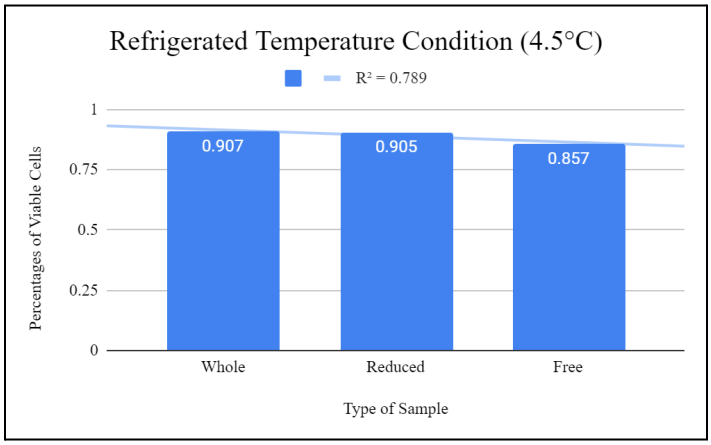
Example: Average estimated cell concentration in whole milk samples 1 and 2 = $\frac{21{,}500{,}000 + 3{,}000{,}000}{2} =$

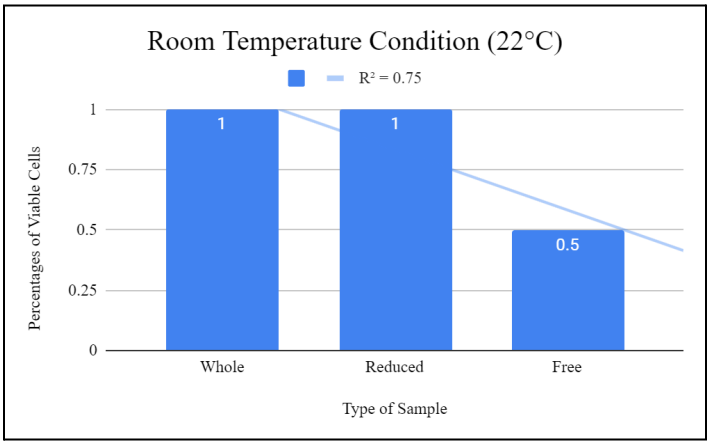
12,250,000

5.1: Graphs

All graphs were created and R^2 values calculated using the 'chart' feature found on Google Sheets.

Graph Type 1: Percentage of Viable Bacteria Cells Across All Milk Samples (both Refrigerated and Room Temperature Conditions)

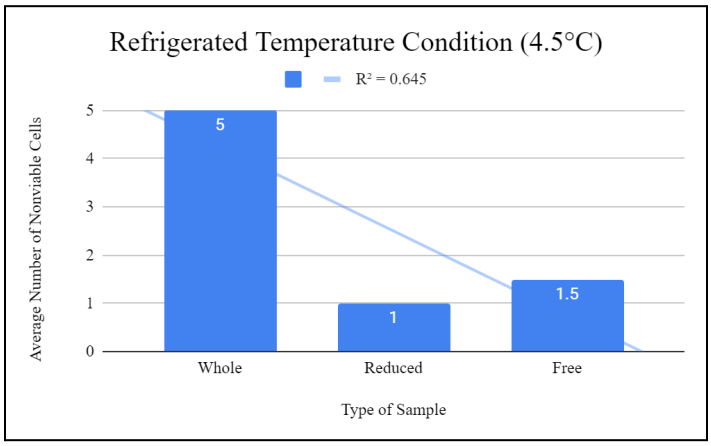


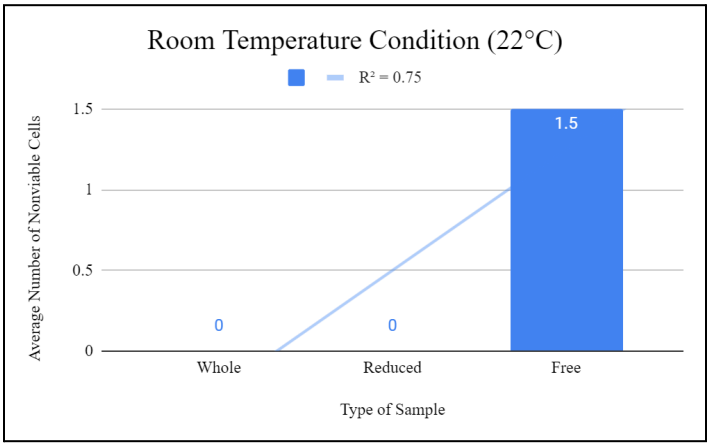


R^2 Value for Refrigerated Temperature Conditions: 0.789

R^2 Value for Room Temperature Conditions: 0.75

Graph Type 2: Average Number of Nonviable Bacteria Cells Across All Milk Samples (both Refrigerated and Room Temperature Conditions)

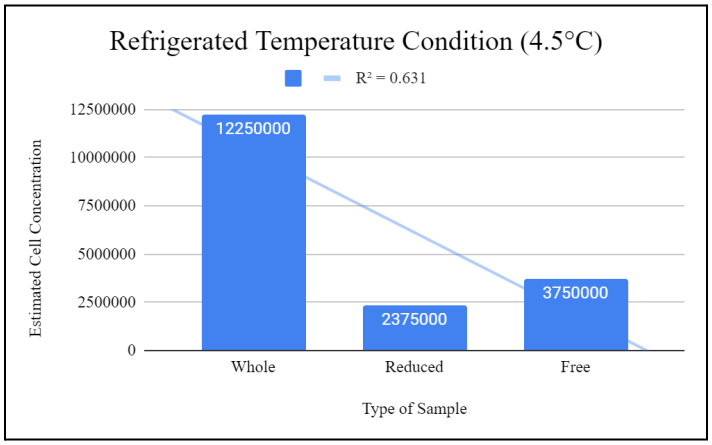


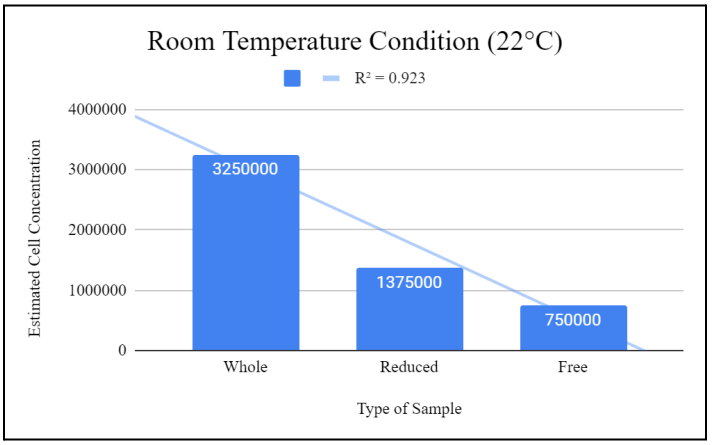


R^2 Value for Refrigerated Temperature Conditions: 0.645

R^2 Value for Room Temperature Conditions: 0.75

Graph Type 3: Estimated Cell Concentration Across All Milk Samples (both Refrigerated and Room Temperature Conditions)





R^2 Value for Refrigerated Temperature Conditions: 0.631

R^2 Value for Room Temperature Conditions: 0.923

5.2: Statistical Analysis

For the percentage of viable bacteria cells found within each solution in the refrigerated temperature condition, a negative linear relationship was observed across the solutions made from the whole, reduced fat, and fat-free milk samples. An R^2 value of 0.789 in this condition indicates that the relationship between the percentages and the type of milk used was strong. A similar relationship was found in the room temperature condition, with an R^2 value of 0.75. This secondary relationship strengthens the overall relationship between fat content and percentage of viable bacteria because a similar relationship exists regardless of temperature, which is an extremely important factor in determining microbial growth. This may mean that a reduction in the amount of fat will generally lead to a decrease in the percentage of viable cells found in the liquid, which could make it safer to consume. This finding supports my hypothesis because a lower percentage of viable cells means there is less living bacteria found in each milk sample as the fat content decreases.

For the average number of nonviable bacteria cells found within each solution in the refrigerated temperature condition, a negative linear relationship was observed across the solutions made from the whole, reduced fat, and fat-free milk samples. An R^2 value of 0.645 in this condition indicates that the relationship between the average number of nonviable cells found and the type of milk used was relatively strong. However, for the room temperature condition, a positive linear relationship was observed across the solutions made from the whole, reduced fat, and the fat-free milk samples, and the average number of nonviable bacteria cells. An R^2 value of 0.75 also indicates that this relationship was strong. In the case of the refrigerated temperature conditions, this relationship does not support my hypothesis because it means that less dead cells were found as the fat content decreased, but this may have been due to improper cleaning procedures taken before counting the number of bacteria cells in the solution made from the first sample of whole milk in the refrigerated condition, spiking the overall averages. In the case of the room temperature conditions, however, this relationship does support my hypothesis because it means that more dead bacteria cells were found as the level of fat content decreased.

Since this had a stronger relationship than the solutions in the refrigerated temperature condition, this may be the more applicable result.

For the estimated viable cell concentration found within each solution in the refrigerated temperature condition, a negative linear relationship was observed across the solutions made from the whole, reduced fat, and fat-free milk samples. An R^2 value of 0.631 in this condition indicates that the relationship between the estimated cell concentration and the type of milk used was relatively strong. A similar yet much stronger relationship was found in the room temperature condition, with an R^2 value of 0.923. This secondary relationship strengthens the overall relationship between fat content and estimated viable cell concentration because a similar relationship exists regardless of temperature, and the fact that the room temperature condition has an even stronger relationship demonstrates the effect that the amount of fat content most likely has on bacterial growth. This may mean that a reduction in the amount of fat will generally lead to a decrease in estimated viable cell concentration, which would indicate that consuming milk with 0% fat is generally safer than 2% or 3.25% milk. This finding supports my hypothesis because these relationships show that more living bacteria cells are found with increasing fat content. However, as mentioned previously, improper cleaning procedures were taken prior to creating one of the solutions, which may have spiked the averages in the refrigerated temperature condition for solutions created from the whole milk samples.

ANOVA Test and Analysis

An ANOVA was conducted to find whether or not there was a statistically significant relationship between fat content and estimated viable cell concentration in milk. A two-factor with replication variant was used due to multiple levels of fat content at different temperatures (two independent variables) and multiple observations (samples) being used for each independent variable.

Processed Data Table 4 (screenshot from Google Sheets): Input Variables (ANOVA Setup)

Sample	Temperature	Fat Content	Bacterial Cell Counts
1	4.5	3.25	21500000
2	4.5	3.25	3000000
3	4.5	2	2750000
4	4.5	2	2000000
5	4.5	0	6000000
6	4.5	0	1500000
7	22	3.25	5000000
8	22	3.25	1500000
9	22	2	2000000
10	22	2	750000
11	22	0	750000
12	22	0	0

Sample number indicates which label was applied to each solution when creating the solutions (not included in ANOVA, simply used for organization).

Temperature was measured in degrees Celsius (°C).

Fat content was measured in percentage (%).

Bacteria cell counts were measured in cells/mL.

Processed Data Table 5 (screenshots from Google Sheets): ANOVA Test (Two-Factor With Replication)

Anova: Two-Factor With Replication								
SUMMARY (p1)	4.5	3.25	Total		SUMMARY (p2)	4.5	3.25	Total
4.5					*22*			
Count	4	4	8		Count	3	3	6
Sum	7.25	13750000	13750007.25		Sum	2	1500000	1500002
Average	1.8125	3437500	1718750.906		Average	0.6666666667	500000	250000.3333
Variance	1.807291667	3098958333333	4704237511162		Variance	1.333333333	187500000000	149999800001
4.5					*Total*			
Count	4	4	8		Count	11	11	
Sum	8.5	10000000	10000008.5		Sum	17.75	25250000	
Average	2.125	2500000	1250001.063		Average	1.613636364	2295454.545	
Variance	2.354166667	2833333333333	2999996964288		Variance	1.904545455	3322727272727	
ANOVA								
Source of Variatio	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*		
Sample	9005216319447	2	4502608159723	3.964463246	0.03995527614	3.633723468		
Columns	3161462497315	1	3161462497315	27.83609284	0.000075338340	4.493998418		
Interaction	3415629326733	2	1707814663367	1.503699239	0.2521032374	3.633723468		
Within	1817187500001	16	1135742187501					
Total	6220734561935	21						

The low sample probability value of about 0.0400 ($p < 0.05$) indicates that there is a significant difference in estimated cell concentration between the different levels of fat content. The low column probability value of about 0.0000753 ($p < 0.05$) suggests that there is a significant difference in the estimated cell concentration between the different temperature conditions. The interaction probability value of about 0.252 ($p > 0.05$) indicates that the relationship between fat content and estimated cell concentration does not significantly depend on the temperature condition, which would suggest that the pattern observed in estimated cell concentration is similar regardless of the temperatures the samples were stored at.

6: Conclusion

Based on the statistical analysis of the results, I have learned that it is evident that the amount of fat content in milk has a significant effect on the microbial growth present in the liquid at both refrigerated and room temperature conditions, suggesting that this relationship is sustained regardless of external conditions.

Firstly, regarding the percentage of viable bacteria cells found in each solution, a consistent negative linear relationship was observed across all solutions in both temperature conditions, indicating that a reduction in fat content generally leads to a decrease in the percentage of viable cells. Since this finding is consistent across

both temperature conditions, it suggests that fat content plays a crucial role in determining microbial growth regardless of temperature.

Likewise, the average number of nonviable cells showed a strong negative linear relationship in the refrigerated condition but a positive linear relationship in the room temperature condition. While the refrigerated condition did not necessarily support the hypothesis likely due to a potential failure in adherence to proper cleaning procedures, the stronger relationship observed at room temperature does support the hypothesis, indicating that lower fat content can correlate with a higher number of nonviable cells.

Moreover, the estimated viable cell concentration demonstrated a negative linear relationship across all solutions in both temperature conditions. This indicates that a reduction in fat content generally leads to a decrease in estimated viable cell concentration, with a much stronger relationship observed at room temperature. This finding suggests that consuming milk with lower fat content, such as fat-free milk rather than reduced fat or whole milk, may be safer for consumption in terms of microbial contamination.

Based on my ANOVA test, with a p-value of about 0.0400 for the sample and 0.0000753 for the column, a significant relationship between fat content and microbial growth is evident, allowing me to sustain my original hypothesis, “Milk samples with higher fat content stored at both room and refrigerated temperature conditions will exhibit a significantly greater amount ($p < 0.05$) of microbial growth than those with lower fat content,” and reject the null hypothesis. Additionally, an insignificant interaction p-value of 0.252 suggests that this effect between fat content and microbial growth may not be significantly impacted by temperature variations, demonstrating that this relationship is consistent in multiple environmental contexts.

To conclude, the findings of my study highlight the importance of fat content in determining microbial growth in milk. Since milk with lower fat content may be safer to consume than milk with higher fat content, these results demonstrate that our food safety and preservation methods may need to be modified in favor of greater refrigeration and other such methods for foods or liquids with higher fat content.

6.1: Strengths

Strength	Explanation/Significance
Controlled Experiment Setup	In my experiment, I tried my best to maintain strict control over variables such as fat content, temperature, and storage duration, which helped to ensure that the results were reliable.
Use of Accurate Measurement Techniques	Using precise equipment, such as micropipettes, a hemocytometer, and a microscope, allowed me to accurately measure bacteria cell counts, which also helped to ensure that the results were reliable.
Replicability	Since I used standardized procedures and equipment and was very precise about each step, this experiment can be replicated and even expanded upon by other researchers to verify the findings.

Statistical Analysis (ANOVA and R^2)	Using an ANOVA test and Pearson's Coefficient of Determination allowed me to assess the significance of differences between fat levels and to draw stronger conclusions.
Safety Measures	Including personal protective equipment (glasses, gloves, and antimicrobial wipes) allowed me to minimize the risk of unintended contamination and exposure to harmful microorganisms, allowing for more reliable results.

6.2: Weaknesses

Weakness	Explanation/Significance	Suggested Improvement
Subjective Cell Counting	Since the methodology of using a hemocytometer to find out the total bacteria cell count within a solution relies primarily on the individual counting of cells by an experimenter, cell counts may be inconsistent, which could lead to potentially dramatic changes in bacteria cell counts depending on the dilution factor.	Including a larger number of experimenters and having each do their own individual cell count can help ensure that the cell counts are not being skewed by subjective interpretations.
Potential Contamination	Regardless of the safety precautions I took during this experiment, there is always a chance for either the solutions or the equipment to be unexpectedly contaminated with bacteria due to the environment, which could have impacted cell counts.	Future variations of this experiment could be conducted in a more controlled environment, such as a clean laboratory space, rather than the potentially bacterially-infected space of my school's biology room.
Inconsistent Timing of Temperature Measurements	Since I only measured the temperatures of the milk samples during the middle of the experiment and not at the beginning or end, there could have been fluctuations in the temperatures that could have dramatically impacted microbial growth, affecting results and further conclusions. However, this may not have been that significant as these samples were well-refrigerated prior to the start	Temperatures in each milk sample can be measured prior to the experiment, multiple times during the experiment (perhaps every day or every two days), and after the experiment to minimize the impact that this factor may have on the results.

	of and during the experiment.	
Lack of pH measurement	Potential of hydrogen levels can play a major role in increasing/decreasing the amount of microbial growth occurring within a nutritive medium, and not measuring them creates a confounding variable that could have easily been accounted for with more planning.	pH levels can be measured every few days (or even every day) during the experiment to determine whether or not they are changing within each milk sample, and thus, whether or not they may be having an impact on the bacteria cell counts.
Inconsistent Hemocytometer Methodology	I used capillary action early on during the bacteria cell counts, but soon switched to using manual droplets of each solution onto the chambers of the hemocytometer due to errors in the capillary action method, which could potentially explain the much higher number of bacteria cells counted in the early solutions (solutions created from the milk in the refrigerated condition) and the later solutions (solutions created from the milk in the room temperature condition).	From the start of an experiment like this, it would be best to define each step to make sure that consistent methodology is used throughout, which will keep results as consistent as possible. In this case, defining whether or not the experimenter will be using a manual droplet or capillary action.
Limited Sample Size	I only used 2 milk samples per fat level and temperature condition (12 total samples), which may reduce the generalizability of the findings and limit the statistical analyses.	Using more samples will allow future research to come to a more generalizable conclusion.
Limited Duration of Study	This experiment only lasted 12 days, which may not have captured long-term microbial growth patterns, limiting the applicability of the findings in real-world situations.	Increasing the duration of the experiment will allow for long-term microbial growth patterns to be measured, allowing for these results to be better applied to future milk storage practices.

6.3: Extensions

Due to the limitations of this experiment, there is a need for further research in this area. One avenue for an extension of this experiment could involve investigating the impact of different types of fat, such as natural versus synthetic fats, on microbial contamination. Additionally, expanding this experiment to include a broader range of temperatures, like extreme heat or cold, could reveal how fat content interacts with temperature to

affect microbial growth, which could potentially add to the pattern discovered in this experiment which is that fat content had a significant impact on microbial growth regardless of refrigerated or room temperatures. Moreover, investigating the role of other constituents of milk, such as proteins or sugars, in microbial growth could offer further insight into future food preservation strategies.

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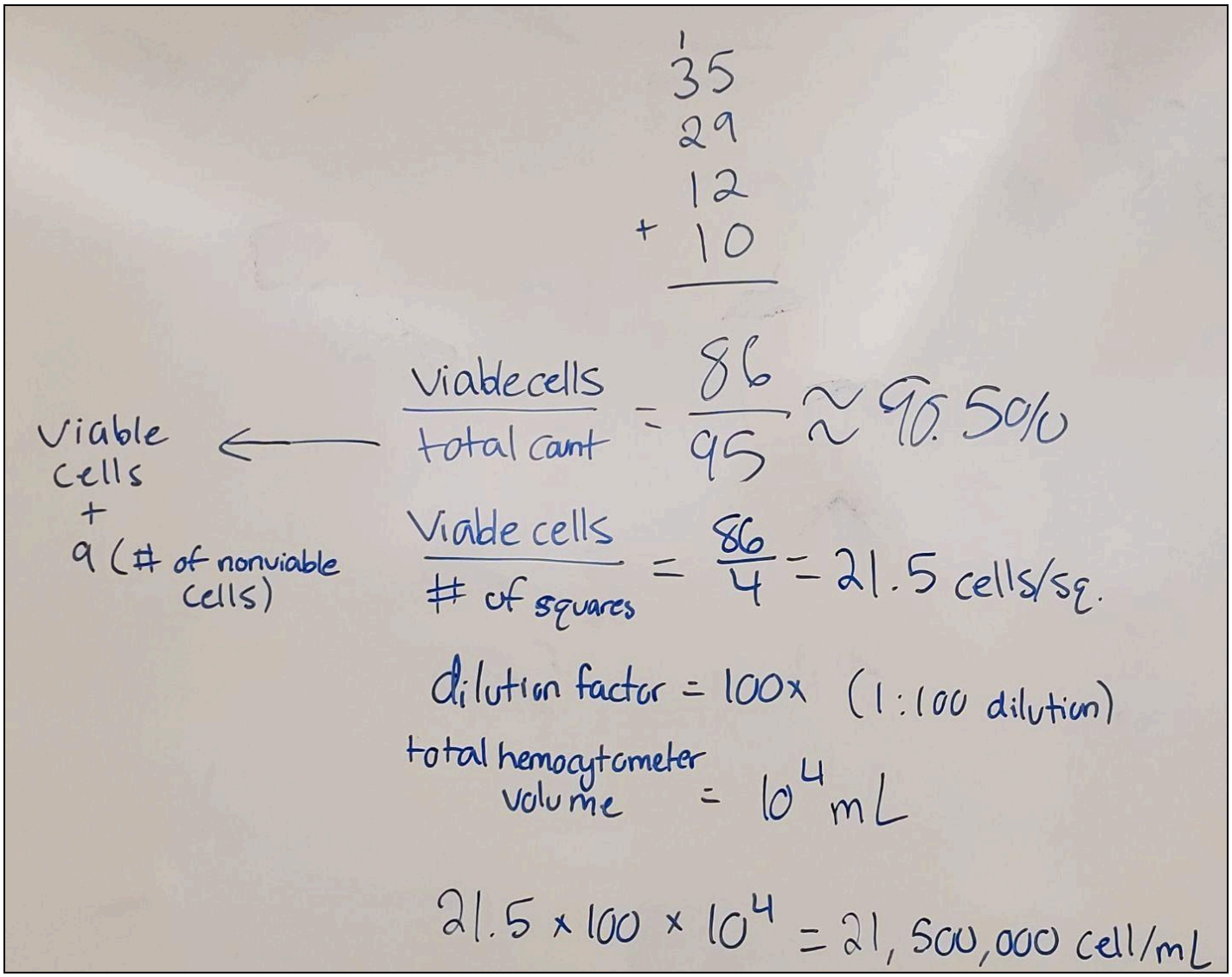
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Appendix

Appendix A. Example of cell concentration calculation



Appendix B. Images of room temperature milk samples (4 days into experiment)



Images of fat-free milk samples



Images of reduced fat milk samples

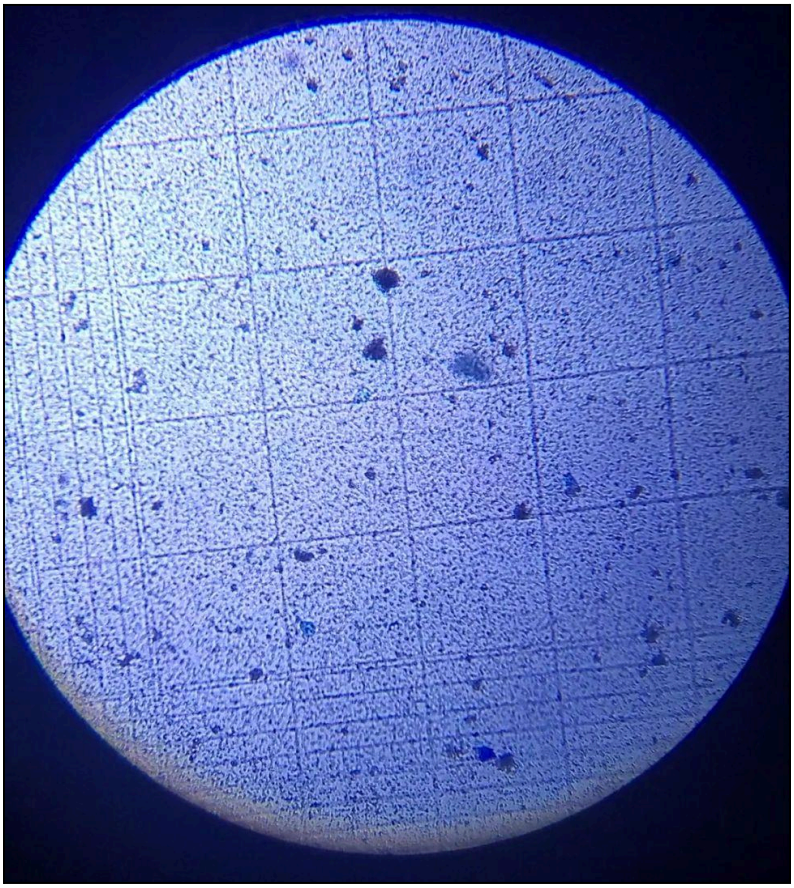


Images of full-fat milk samples

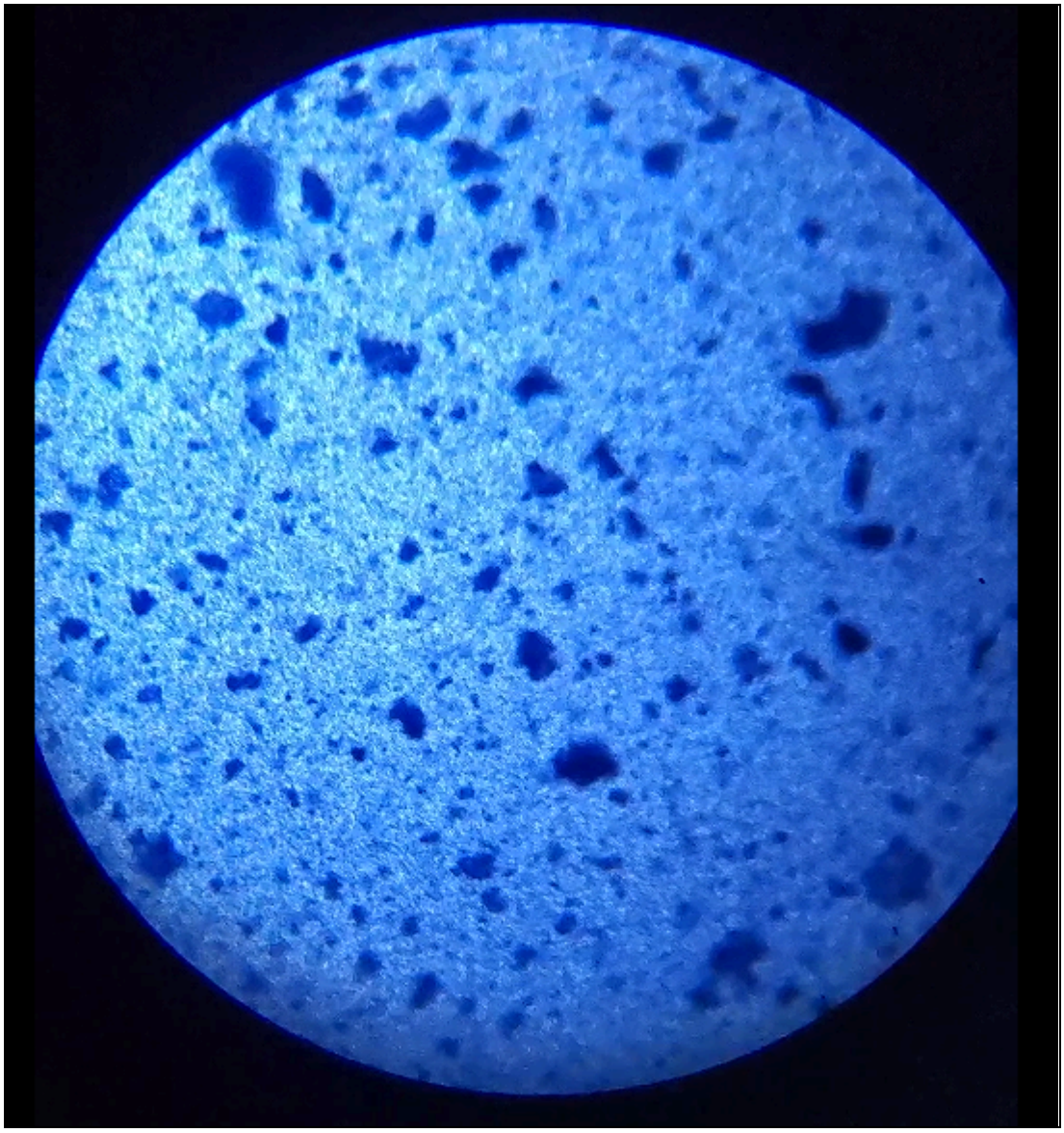
Appendix C. Solution 1 from sample 1 of the 2% milk in the refrigerated condition (lighter blue color)



Appendix D. Example hemocytometer image of one grid of 4.5°C Sample 1 (1:100 dilution)



Appendix E. Example hemocytometer image of one grid of a 1:10 dilution



Appendix F. Image of microscope setup for hemocytometer cell count

